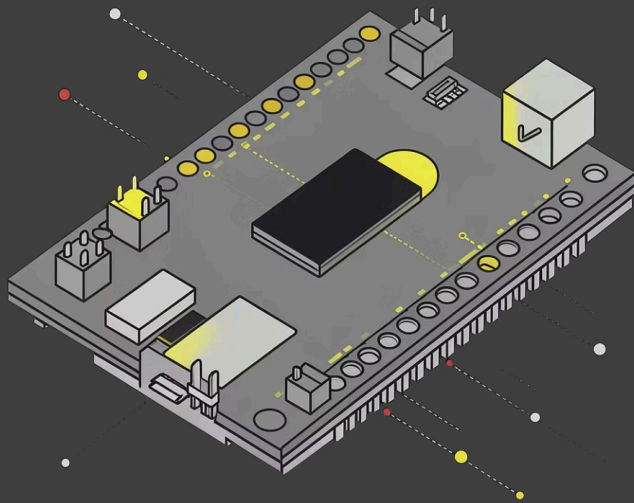
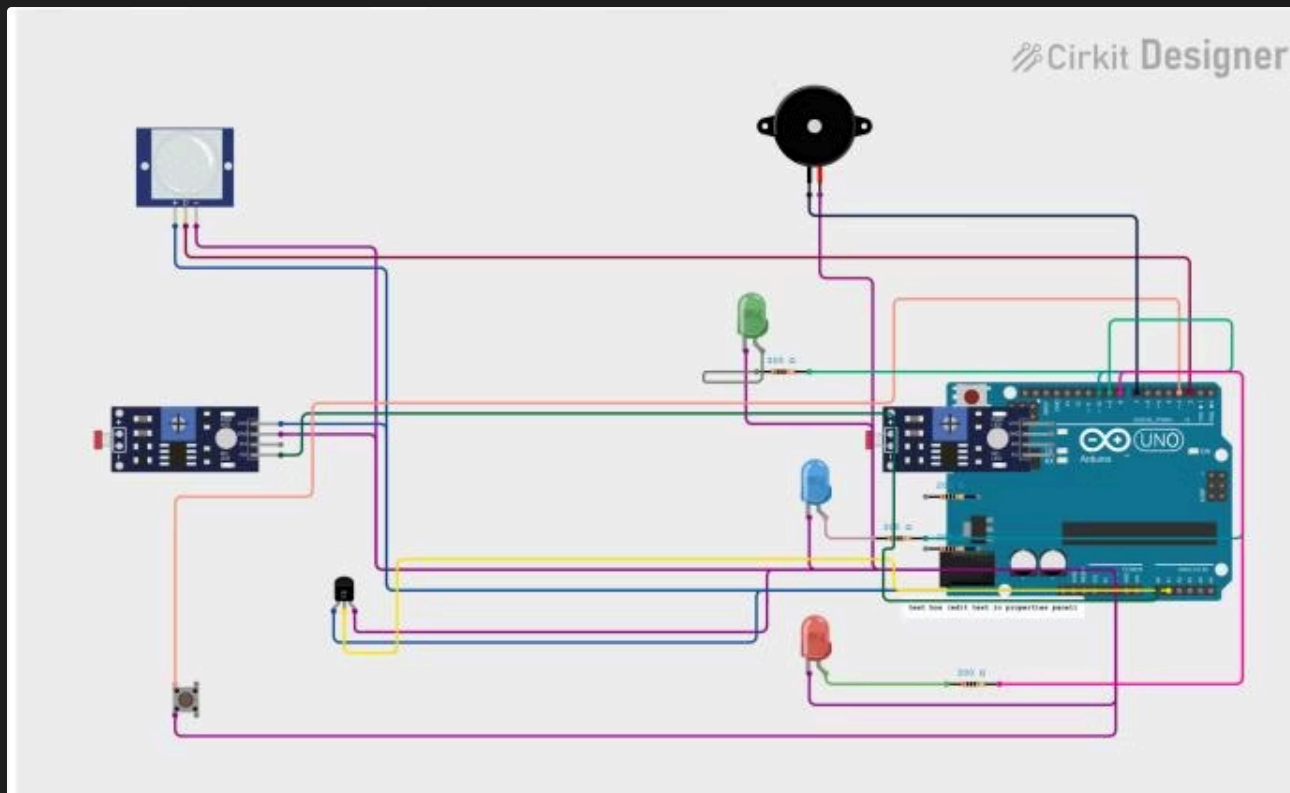




# Arduino Automation Projects

Smart Home • Conveyor Systems • Packaging Automation



# Smart Home Lighting System



## Motion Lighting

Activates when room is **dark AND motion detected**



## Temp Monitor

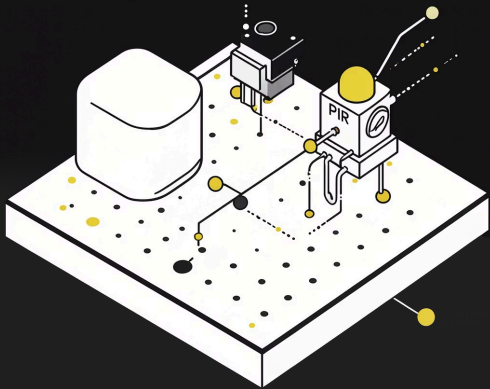
Fan indicator on above **30°C**



## Security Alarm

Buzzer fires if **door opens during motion**

# Smart Home: Key Components



## Sensors

PIR (motion) · LDR  
(light) · LM35 (temp)  
· Door switch

## Outputs

Light LED · Fan LED  
· Alarm LED · Buzzer

## Brain

Arduino UNO —  
real-time sensor  
fusion

# Smart Home: Complete Code

```
// Smart Home Automation - Arduino UNO
```

```
const byte PIR_PIN  = 2;
const byte DOOR_PIN = 3;
const byte ALARM_LED = 8;
const byte LIGHT_LED = 9;
const byte FAN_LED  = 10;
const byte BUZZER_PIN = 7;
const byte LDR_PIN  = A0;
const byte TEMP_PIN  = A1;
```

```
void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(DOOR_PIN, INPUT_PULLUP);
  pinMode(ALARM_LED, OUTPUT);
  pinMode(LIGHT_LED, OUTPUT);
  pinMode(FAN_LED, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  Serial.begin(9600);
}
```

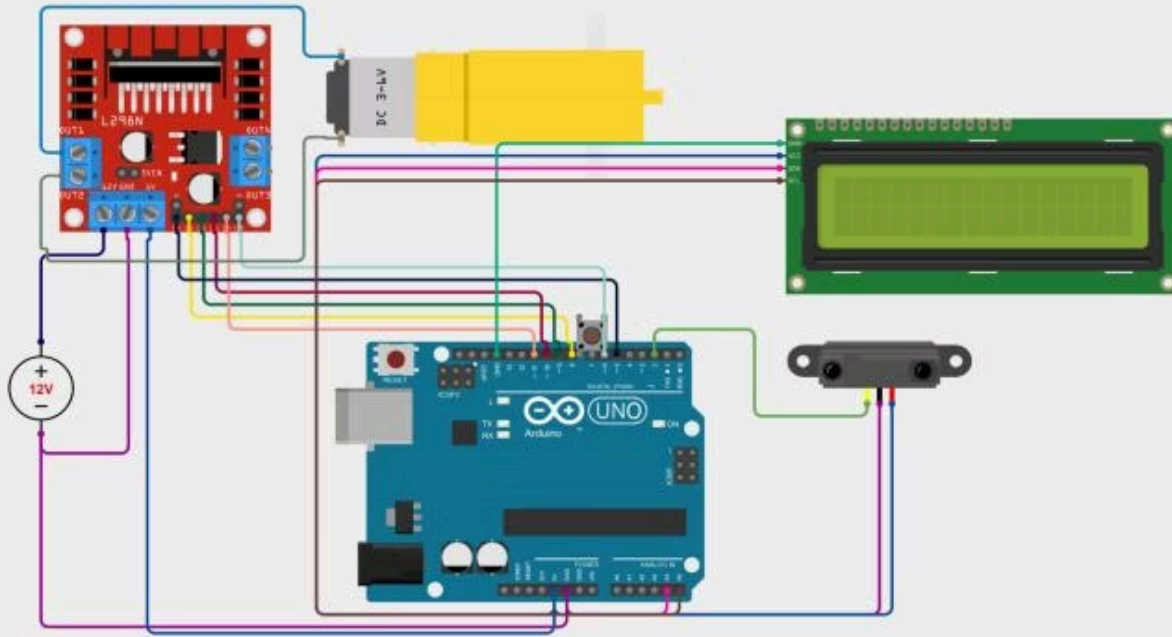
```
void loop() {
  int lightLevel = analogRead(LDR_PIN);
  int tempRaw = analogRead(TEMP_PIN);
  float temperatureC = tempRaw*(5.0/1023.0)*100.0;
  bool motionDetected = digitalRead(PIR_PIN)==HIGH;
  bool doorOpen = digitalRead(DOOR_PIN)==LOW;
```

```
  // Light: dark + motion
  digitalWrite(LIGHT_LED,
    (lightLevel < 500 && motionDetected)
    ? HIGH : LOW);
```

```
  // Fan: temp > 30°C
  digitalWrite(FAN_LED,
    temperatureC > 30.0 ? HIGH : LOW);
```

```
  // Alarm: door open + motion
  bool alarm = doorOpen && motionDetected;
  digitalWrite(ALARM_LED, alarm ? HIGH : LOW);
  if (alarm) tone(BUZZER_PIN, 2000);
  else noTone(BUZZER_PIN);
```

```
  delay(200);
}
```



# Conveyor Belt with Sensors

## Item Counting

IR sensor detects each object **once per passage**

## Live Display

16x2 I2C LCD shows **real-time count**

## Continuous Belt

DC motor at **speed 180** PWM units

## Visual Feedback

LED lights up on **each detection event**

# Conveyor Belt: Complete Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
const int IR_SENSOR_PIN = 2;
const int MOTOR_IN1 = 8;
const int MOTOR_IN2 = 9;
const int MOTOR_ENA = 10;
const int LED_PIN = 13;
const int BELT_SPEED = 180;
int objectCount = 0;
bool isObjectDetected = false;

void setup() {
  lcd.init();
  lcd.backlight();
  pinMode(IR_SENSOR_PIN, INPUT);
  pinMode(MOTOR_IN1, OUTPUT);
  pinMode(MOTOR_IN2, OUTPUT);
  pinMode(MOTOR_ENA, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  lcd.print("Conveyor System");
  runBeltForward();
}
```

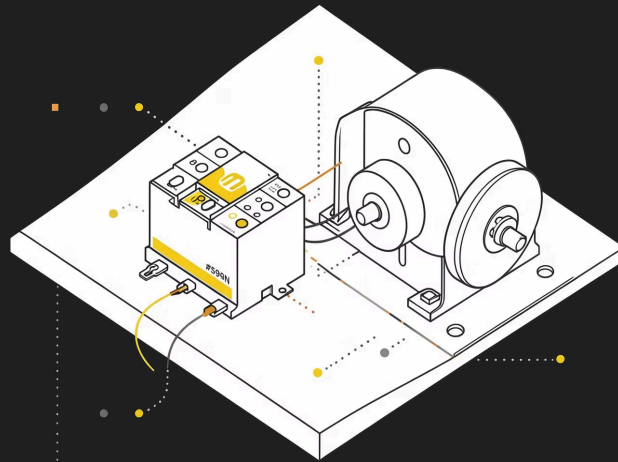
```
void loop() {
  int sensorState = digitalRead(IR_SENSOR_PIN);

  if (sensorState == LOW && !isObjectDetected) {
    objectCount++;
    isObjectDetected = true;
    digitalWrite(LED_PIN, HIGH);
    lcd.setCursor(12, 1);
    lcd.print(objectCount);
    lcd.print(" ");
  }
  else if (sensorState == HIGH && isObjectDetected){
    isObjectDetected = false;
    digitalWrite(LED_PIN, LOW);
  }

  runBeltForward();
  delay(20);
}

void runBeltForward() {
  digitalWrite(MOTOR_IN1, HIGH);
  digitalWrite(MOTOR_IN2, LOW);
  analogWrite(MOTOR_ENA, BELT_SPEED);
}
```

# Conveyor System: Hardware Stack



01

---

## Detection

IR sensor on **pin 2**

02

---

## Motor Control

L298N driver — PWM speed regulation on **pin 10**

03

---

## Display

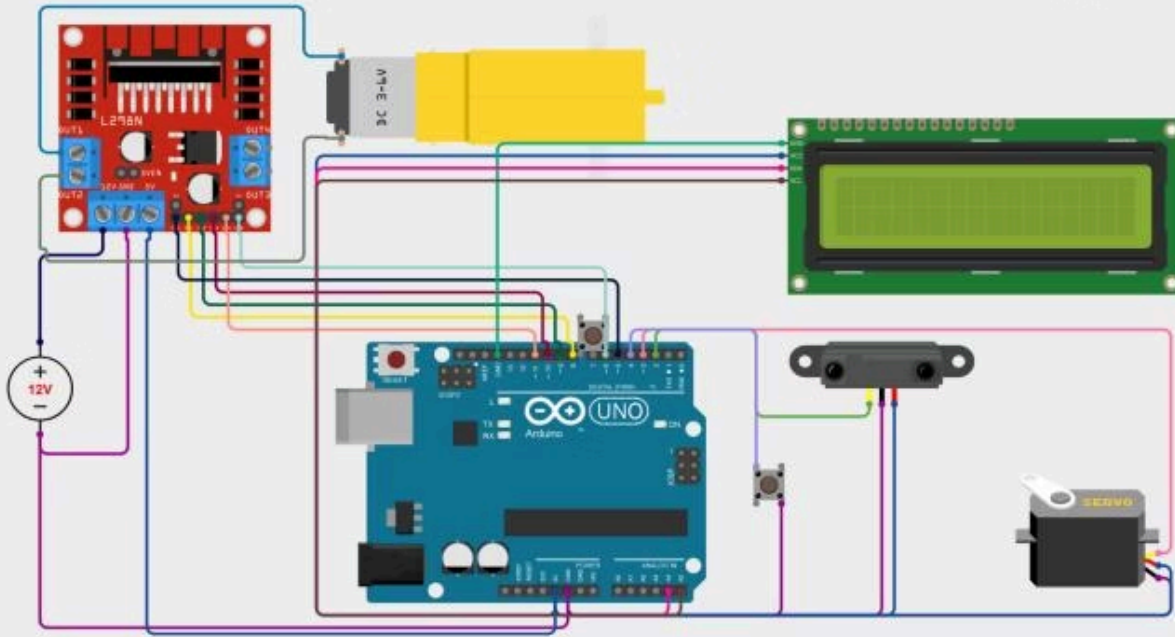
I2C LCD at address **0x27** — live item count

04

---

## Feedback

LED indicator on **pin 13**



# Automated Packaging System

1

 Detect

IR sensor spots incoming item

2

 Stop

Belt halts for packing

3

 Pack

Servo pushes at **90°**

4

 Resume

Belt restarts automatically



# Packaging System: Complete Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo packServo;

const int irSensorPin = 2;
const int servoPin    = 3;
const int buttonPin   = 4;
const int motorPin1   = 8;
const int motorPin2   = 9;

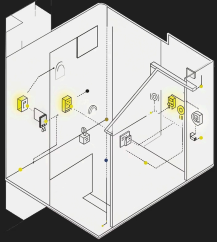
int itemCount    = 0;
bool systemActive = true;
bool lastButtonState= HIGH;

void setup() {
  pinMode(irSensorPin, INPUT);
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);
  packServo.attach(servoPin);
  packServo.write(0);
  lcd.init();
  lcd.backlight();
  lcd.print("PACKAGING SYSTEM");
}
```

```
void loop() {
  // Button toggle start/pause
  bool btnState = digitalRead(buttonPin);
  if (btnState==LOW && lastButtonState==HIGH){
    systemActive = !systemActive;
    delay(50);
  }
  lastButtonState = btnState;

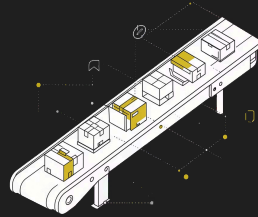
  if (systemActive) {
    runMotor();
    if (digitalRead(irSensorPin)==LOW) {
      stopMotor();
      delay(300);
      // Servo pack action
      packServo.write(90);
      delay(600);
      packServo.write(0);
      delay(300);
      itemCount++;
      lcd.setCursor(0,1);
      lcd.print("Packed: ");
      lcd.print(itemCount);
      runMotor();
    }
  } else {
    stopMotor();
  }
  delay(20);
}
```

# Project Comparison at a Glance



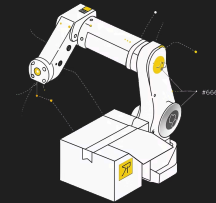
## Smart Home

PIR · LDR · LM35 · Door switch  
Arduino UNO · Buzzer · LEDs



## Conveyor Belt

IR sensor · L298N · DC motor  
I2C LCD · PWM speed control



## Packaging System

IR sensor · Servo motor · I2C LCD  
Push button · Auto workflow

# Build. Automate. Innovate.

## ✓ 3 Projects

Complete working code  
included

## ⚡ Real Hardware

Drop-in ready for Arduino UNO

## 🚀 Scalable

Extend with WiFi, cloud & more

